3D Computer Vision

Project #3: Point Cloud Registration & SLAM Basics

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LLM Usage Policy Notice
We have used GPT to help refine this project description! You may use AI tools for assistance as well, but you must be able to explain every line of your code and may be asked to implement parts from scratch in a face-to-face meeting with the TAs.

Objective

The goal of this project is to understand and implement the basics of **Simultaneous Localization and Mapping (SLAM)** using the **Iterative Closest Point (ICP)** algorithm for point cloud registration.

In practical terms, SLAM involves building a map of an unknown environment while simultaneously tracking the position of the sensor (in this case, a vehicle equipped with a LIDAR scanner). This project emphasizes the registration (alignment) of 3D point clouds collected over time, and how this registration process contributes to estimating both the environment and the trajectory of the robot.

You will use a subset of the **KITTI dataset**—specifically, a portion of one sequence containing a series of point clouds captured by a LIDAR sensor mounted on a moving vehicle. These point clouds represent snapshots of the surrounding environment at successive time steps.

The main tasks in this project include:

- Applying the ICP algorithm to register pairs of point clouds.
- Incrementally fusing multiple point clouds to create a unified 3D map.
- Estimating the robot's trajectory from the transformations obtained during registration.
- Evaluating the estimated trajectory by comparing it with groundtruth using the **Final Displacement Error (FDE)** and **Average Displacement Error (ADE)**.

Figure 1: Map of Sequence 00 of KITTI Dataset.

Data and Tools

You are provided with a subset of point clouds extracted from a single sequence of the **KITTI dataset**, which was collected using a LIDAR-equipped vehicle. This subset contains a portion of the full sequence and represents the environment at successive time steps as the vehicle moves forward. Although it is not the complete sequence, it still captures enough temporal and spatial overlap to perform reliable point cloud registration and path estimation.

Implement in Python with appropriate libraries (e.g., Open3D, NumPy, scipy). You will need to implement or use ICP for point cloud registration and trajectory estimation.

1. **Point Cloud Registration Using ICP**

   (a) **Select Two Point Clouds:** Select two point clouds that are no more than 10 frames apart in the dataset. These point clouds should represent the same region of the environment at different times. (Selecting frames close in time ensures sufficient overlap between scans, which is necessary for the ICP algorithm to find correspondences and converge to a good alignment.)

   (b) **Visualize Unregistered Point Clouds:** Before applying ICP, plot and visualize the two selected point clouds together in a single 3D plot. Use different colors to distinguish them clearly, showing their initial, unaligned state.

(c) **Apply ICP Algorithm:** Use the Iterative Closest Point (ICP) algorithm to register the two point clouds by estimating the best rigid transformation that aligns one cloud to the other.

(d) **Visualize Aligned Point Clouds:** After applying ICP, plot the transformed source point cloud together with the target point cloud to illustrate the alignment result.

2. **Incremental Registration of Multiple Point Clouds Using ICP**

   (a) **Use the Result from Previous Part:** Use the aligned pair from the previous part as the initial base map and iteratively register subsequent point clouds to this growing map using ICP.

   (b) **Visualize the Final Map:** Combine and plot all the registered point clouds to visualize the complete unified 3D map of the environment. *(The final map should reflect the trajectory and structure of the scene as captured across all scans.)*

3. **Path Evaluation Using FDE and ADE**

   (a) **Visualize Paths:** Plot the estimated robot trajectory over time alongside the groundtruth trajectory in the 2D (x, y) plane.

   (b) **Calculate Final Displacement Error (FDE):** Compute the Euclidean distance between the final estimated position and the final groundtruth position:

$$\text{FDE} = \sqrt{(x_{\text{est}}(T) - x_{\text{gt}}(T))^2 + (y_{\text{est}}(T) - y_{\text{gt}}(T))^2}$$

   (c) **Calculate Average Displacement Error (ADE):** Compute the average Euclidean distance between estimated and groundtruth positions over all time steps:

$$\text{ADE} = \frac{1}{T} \sum_{t=1}^{T} \sqrt{(x_{\text{est}}(t) - x_{\text{gt}}(t))^2 + (y_{\text{est}}(t) - y_{\text{gt}}(t))^2}$$

   (d) **Analysis and Discussion:** Interpret the FDE and ADE results, discussing the performance of ICP and relating these metrics to the visual map and registration quality.

Submission

Submit a concise PDF report (3–5 pages) containing: a description of the methods and equations you used, clear figures for each stage (unregistered point clouds, aligned pairs, incremental registration results, final unified map, trajectory visualization), numerical results (FDE, ADE, computation times), and thorough discussion. Provide runnable code (Python) and a script to reproduce all figures.